Mean Variance Portfolio Optimisation

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Project Overview

Project Description

This project explores portfolio optimisation and risk analysis using historical asset prices. It begins by collecting historical daily price data, computing daily returns, and annualising them to derive the mean returns vector and covariance of returns matrix. Using these inputs, an efficient frontier is plotted to determine efficient portfolio weight combinations. A Monte Carlo simulation is employed to generate potential asset returns scenarios, assuming asset returns follow a multivariate normal distribution, leading to the calculation of risk measures such as Value at Risk (VaR) and Expected Shortfall (ES). Efficient portfolios are then compared across these risk measures, offering an integrated assessment of efficiency and risk-adjusted performance. Statistical analysis, optimisation techniques, and simulation-based risk evaluation are implemented to demonstrate practical applications of mean-variance portfolio theory.

```r
library(quantmod)
library(PerformanceAnalytics)
library(quadprog)
library(MASS)
library(ggplot2)
library(knitr)
```

Data Overview

A set of five assets (AAPL, XOM, GLD, TLT, WMT) were selected to construct a portfolio across distinct economic sectors, asset classes and risk factor exposures, enhancing diversification within the Mean-Variance Portfolio Theory (MVPT) framework.

AAPL	Apple Inc.
XOM	Exxon Mobil Corporation
GLD	Gold Futures
TLT	20-Yr Treasury Bond ETF
WMT	Walmart Inc.

Historical Prices

Historical daily prices for each of the assets are stored as a time series object (xts) in R, and any missing values have been omitted. Daily prices from 01/01/2020 to 06/02/2026 are considered.

Historical Prices Data:

The first few rows are displayed to aid in understanding the daily prices data.

```
##                AAPL      XOM    GLD      TLT      WMT
## 2020-01-02 72.40051 53.30640 143.95 115.0038 36.43383
## 2020-01-03 71.69665 52.87785 145.86 116.7749 36.11219
## 2020-01-06 72.26793 53.28386 147.39 116.1118 36.03868
## 2020-01-07 71.92805 52.84777 147.97 115.5410 35.70478
## 2020-01-08 73.08512 52.05082 146.86 114.7772 35.58226
## 2020-01-09 74.63750 52.44931 146.03 115.1801 35.94985
```

Historical Prices Summary Statistics:

Quartiles are reported to exhibit the skewness of the daily prices data.

AAPL	XOM	GLD	TLT	WMT
Min. : 54.21	Min. : 23.99	Min. :138.0	Min. : 75.09	Min. : 31.87
1st Qu.:131.86	1st Qu.: 50.31	1st Qu.:167.9	1st Qu.: 86.77	1st Qu.: 43.72
Median :164.33	Median : 94.46	Median :179.1	Median : 93.12	Median : 47.36
Mean :166.28	Mean : 80.48	Mean :207.3	Mean :103.73	Mean : 58.58
3rd Qu.:200.77	3rd Qu.:105.56	3rd Qu.:223.9	3rd Qu.:123.64	3rd Qu.: 68.77
Max. :285.92	Max. :146.61	Max. :495.9	Max. :145.38	Max. :128.00

Historical Returns

From historical daily prices, historical daily returns are simply computed. These returns are then annualised by assuming 252 trading days per annum, and a mean annual returns vector and a covariance of annual returns matrix are produced.

Historical Returns Data:

The first few rows are displayed to aid in understanding of the daily returns data.

```
##                    AAPL          XOM          GLD          TLT          WMT
## 2020-01-03 -0.009721720 -0.008039386  0.013268522  0.015400051 -0.008828058
## 2020-01-06  0.007968035  0.007678184  0.010489502 -0.005678261 -0.002035684
## 2020-01-07 -0.004703081 -0.008184110  0.003935151 -0.004915630 -0.009264952
## 2020-01-08  0.016086548 -0.015080274 -0.007501525 -0.006610717 -0.003431592
## 2020-01-09  0.021240654  0.007655879 -0.005651653  0.003510018  0.010330762
## 2020-01-10  0.002260785 -0.008887816  0.006026192  0.008890707 -0.008350667
```

Historical Returns Summary Statistics:

Quartiles are reported to exhibit the skewness of the daily returns data.

AAPL	XOM	GLD	TLT	WMT
Min. :-0.128647	Min. :-0.1222480	Min. :-0.1027425	Min. :-0.0666830	Min. :-0.1137576
1st Qu.:-0.008158	1st Qu.:-0.0101511	1st Qu.:-0.0046624	1st Qu.:-0.0067359	1st Qu.:-0.0060248
Median : 0.001097	Median : 0.0005833	Median : 0.0009103	Median :-0.0001096	Median : 0.0007936
Mean : 0.001072	Mean : 0.0008676	Mean : 0.0007927	Mean :-0.0001213	Mean : 0.0009204
3rd Qu.: 0.011240	3rd Qu.: 0.0118222	3rd Qu.: 0.0064962	3rd Qu.: 0.0060406	3rd Qu.: 0.0077832
Max. : 0.153288	Max. : 0.1268679	Max. : 0.0635872	Max. : 0.0751955	Max. : 0.1170848

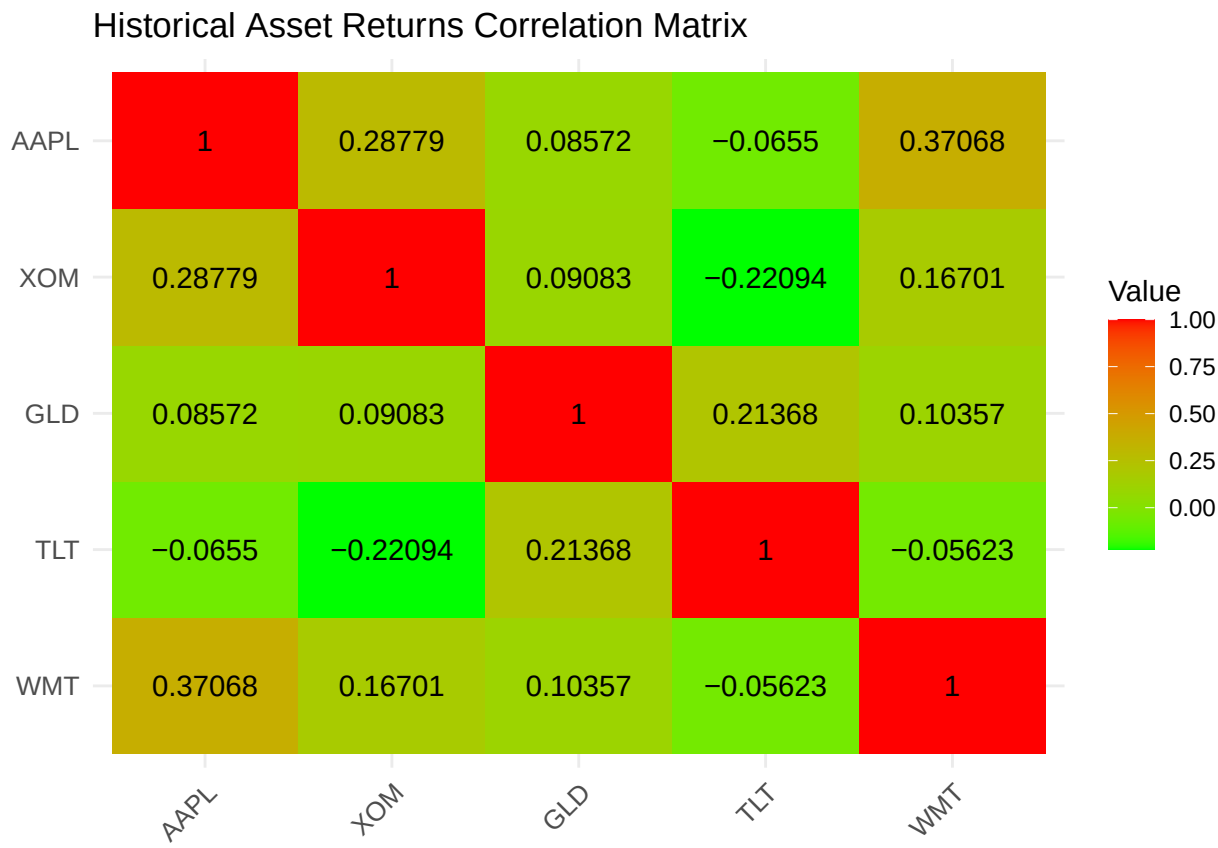
Mean Annual Returns Vector (Mu)

AAPL	0.2702148
XOM	0.2186439
GLD	0.1997615
TLT	-0.0305788
WMT	0.2319489

Covariance of Annual Returns Matrix (Sigma)

	AAPL	XOM	GLD	TLT	WMT
AAPL	0.1005009	0.0299112	0.0047097	-0.0035318	0.0270352
XOM	0.0299112	0.1074879	0.0051612	-0.0123207	0.0125971
GLD	0.0047097	0.0051612	0.0300387	0.0062994	0.0041299
TLT	-0.0035318	-0.0123207	0.0062994	0.0289310	-0.0022002
WMT	0.0270352	0.0125971	0.0041299	-0.0022002	0.0529289

Correlation of Returns Matrix



Among the assets, there is low correlation and in some cases negative correlation. This is desirable as it improves portfolio efficiency (reduces total volatility without necessarily reducing expected return).

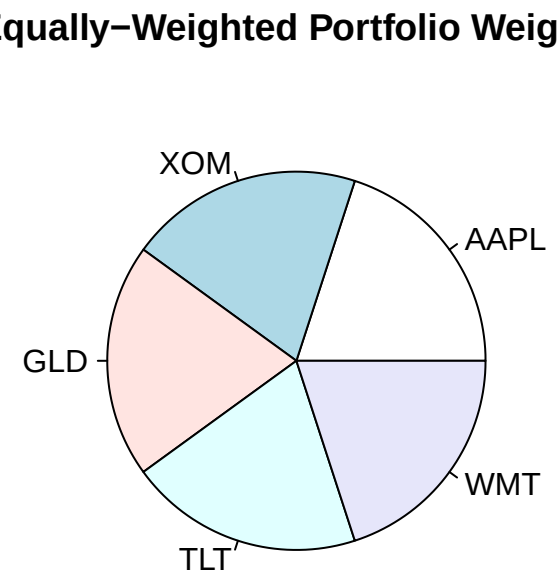
Example Portfolio

Using the mean annual returns vector, mu (mu_annual), and the covariance of annual returns matrix, sigma (cov_annual), the expected return and variance of any portfolio of the selected assets can be determined. This will be demonstrated with the equally-weighted porfolio (EW) for convenience - this portfolio has all positive weights, however it is assumed in general that short selling (negative weights) is allowed.

```
ew <- rep(1/N, N)
mu_ew <- ew%*%mu_annual
var_ew <- ew%*%cov_annual%*%ew
```

```
## [1] "Expected Return (EW): 0.177998055454727"
```

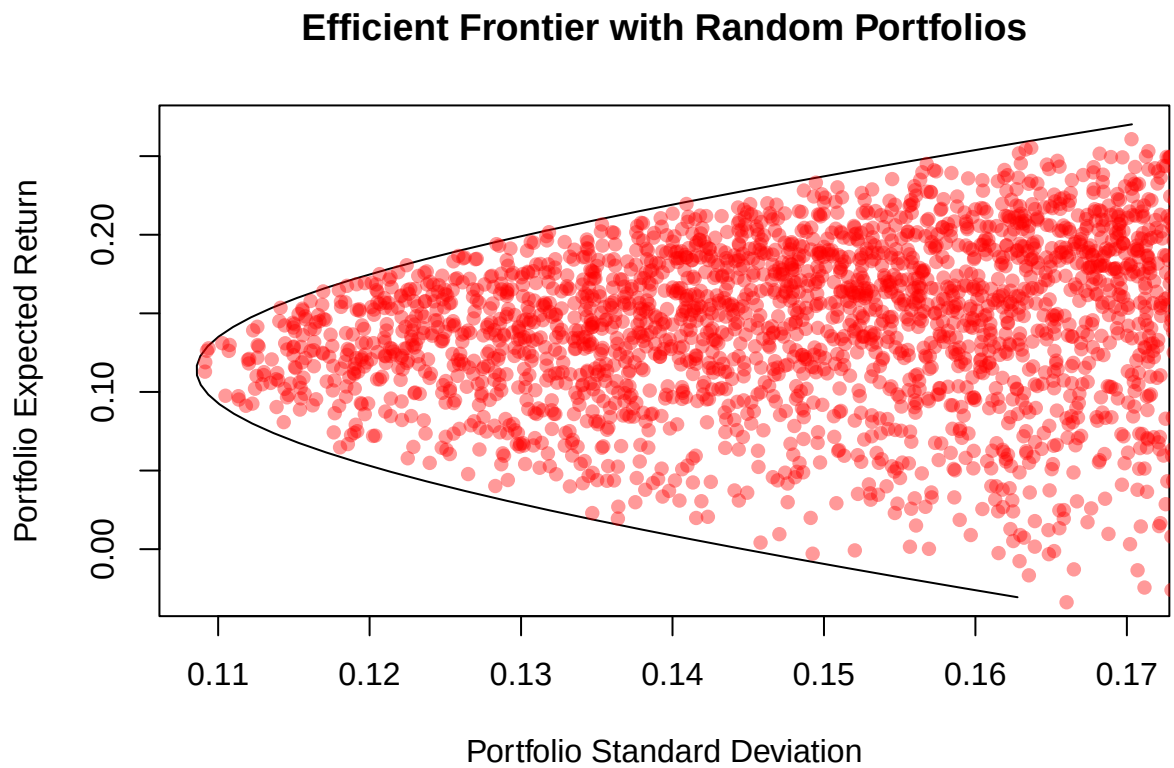
```
## [1] "Variance of Return (EW): 0.0185387782634257"
```



Efficient Frontier

Tracing the Efficient Frontier (Constrained Quadratic Optimisation)

The efficient frontier is plotted by determining the minimum variance for every expected return value between the minimim and maximum expected returns found in the mean annual returns vector, mu, as these are the minimum and maximum expected returns achievable by any portfolio of the assets. The continuous line is discretised into fifty points and the minimum variance for each expected return is determined using constrained quadratic optimisation and the quadprog library. The curve generated is a hyperbola in Expected Return-Standard Deviation (E-sigma) space. By plotting randomly selected portfolios, it can be seen that the curve partitions the space into attainable but inefficient (right), and unattainable portfolios (left). The efficient frontier is the top arc of the hyperbola.



Global Minimum Variance Portfolio (GMV)

The weights of the GMV are given by C^(-1)1 / t(1)C^(-1)1, where C is the covariance of annual returns matrix, C^(-1) is its inverse and 1 is a vector of ones. C is necessarily positive semi-definite, however for it to be positive definite and hence necessarily invertible, its eigenvalues must be positive, which is the case here.

Eigenvalues of C (Sigma):

eigenvalues
0.1442528
0.0783569
0.0410094
0.0351791
0.0210893

All of the eigenvalues of the covariance of annual returns matrix (C) are positive and hence C is positive definite and invertible.

GMV Portfolio Weights:

	weights
AAPL	0.0395864
XOM	0.1141041
GLD	0.2550660
TLT	0.4184382
WMT	0.1728053

GMV Portfolio Mean and Variance:

```
## [1] "GMV Expected Returns: 0.113884026250005"
```

```
## [1] "GMV Variance: 0.01178674685982"
```

Efficient Portfolios

Of the portfolios used in the discretised curve in Expected Return-Standard Deviation Space, only those that are in the upper arc of the curve are efficient. Thus only the portfolios whose expected return exceded the GMV portfolio's expected return are efficient. These efficient portfolios are stored in the dataframe, eff_ports. w_i is the weight corresponding to asset i.

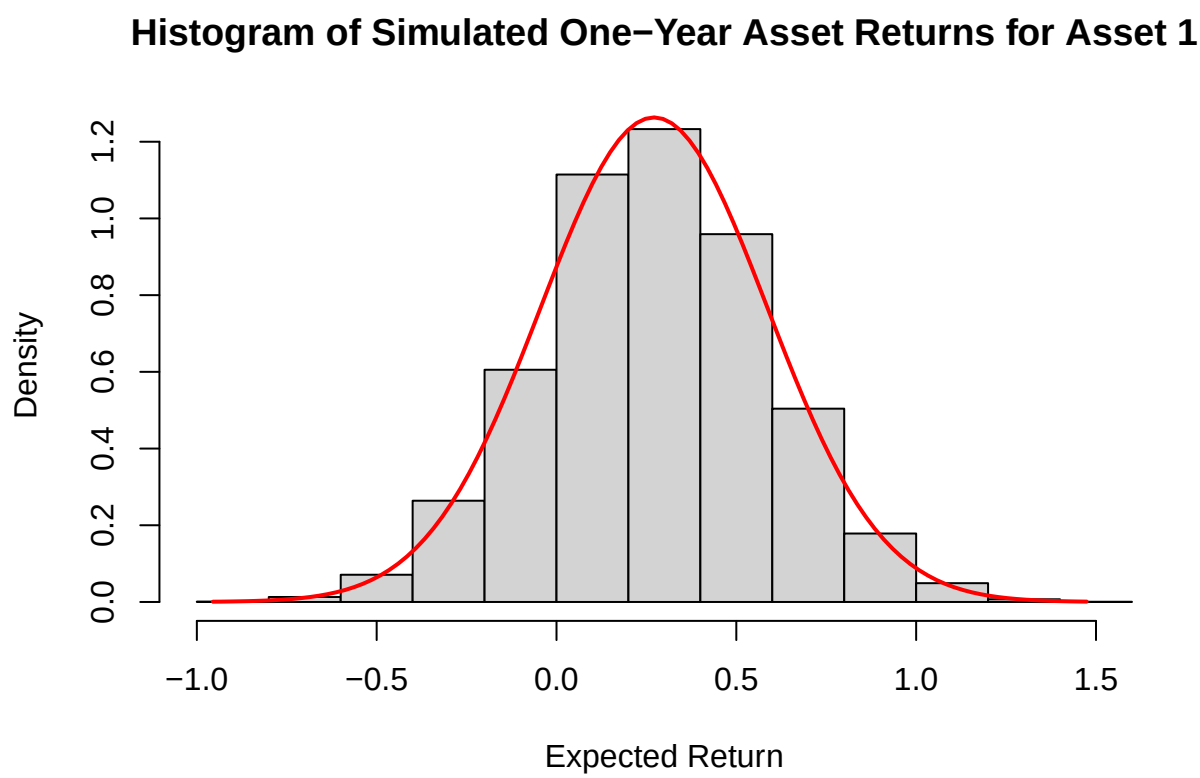
	mu	sd	w1	w2	w3	w4	w5
25	0.1167487	0.1085934	0.0414193	0.1135638	0.2627072	0.4068842	0.1754255
26	0.1228873	0.1088296	0.0453472	0.1124059	0.2790817	0.3821249	0.1810403
27	0.1290259	0.1093086	0.0492750	0.1112480	0.2954561	0.3573657	0.1866552
28	0.1351646	0.1100272	0.0532028	0.1100901	0.3118305	0.3326065	0.1922701
29	0.1413032	0.1109807	0.0571306	0.1089322	0.3282050	0.3078472	0.1978850
30	0.1474419	0.1121631	0.0610585	0.1077743	0.3445794	0.2830880	0.2034998
31	0.1535805	0.1135674	0.0649863	0.1066164	0.3609539	0.2583287	0.2091147
32	0.1597192	0.1151854	0.0689141	0.1054585	0.3773283	0.2335695	0.2147296
33	0.1658578	0.1170082	0.0728419	0.1043006	0.3937028	0.2088103	0.2203445
34	0.1719964	0.1190264	0.0767698	0.1031427	0.4100772	0.1840510	0.2259593
35	0.1781351	0.1212302	0.0806976	0.1019848	0.4264517	0.1592918	0.2315742
36	0.1842737	0.1236098	0.0846254	0.1008268	0.4428261	0.1345325	0.2371891
37	0.1904124	0.1261552	0.0885532	0.0996689	0.4592005	0.1097733	0.2428040
38	0.1965510	0.1288566	0.0924811	0.0985110	0.4755750	0.0850140	0.2484188
39	0.2026897	0.1317043	0.0964089	0.0973531	0.4919494	0.0602548	0.2540337
40	0.2088283	0.1346890	0.1003367	0.0961952	0.5083239	0.0354956	0.2596486
41	0.2149670	0.1378020	0.1042646	0.0950373	0.5246983	0.0107363	0.2652635
42	0.2211056	0.1410346	0.1081924	0.0938794	0.5410728	-0.0140229	0.2708783
43	0.2272442	0.1443789	0.1121202	0.0927215	0.5574472	-0.0387822	0.2764932
44	0.2333829	0.1478273	0.1160480	0.0915636	0.5738217	-0.0635414	0.2821081
45	0.2395215	0.1513726	0.1199759	0.0904057	0.5901961	-0.0883006	0.2877230
46	0.2456602	0.1550082	0.1239037	0.0892478	0.6065705	-0.1130599	0.2933378
47	0.2517988	0.1587279	0.1278315	0.0880899	0.6229450	-0.1378191	0.2989527
48	0.2579375	0.1625260	0.1317593	0.0869320	0.6393194	-0.1625784	0.3045676
49	0.2640761	0.1663970	0.1356872	0.0857741	0.6556939	-0.1873376	0.3101825
50	0.2702148	0.1703360	0.1396150	0.0846162	0.6720683	-0.2120969	0.3157973

Monte Carlo Simulation of Asset Returns

Monte Carlo simulation is used to create random possible asset return scenarios for the five assets where the time horizon is one year. It is assumed that annual assets returns follow a multivariate normal distribution with the mean of annual returns vector and the covariance of annual returns matrix as parameters. 10000 such scenarios are generated, of which the first few are displayed.

AAPL	XOM	GLD	TLT	WMT
0.1449487	0.6096502	0.1590293	0.0436751	0.2908557
0.3621240	0.1723467	0.0325358	0.0414156	0.0222251
0.2913273	0.1715984	0.0828000	-0.1272273	0.3887573
0.0245046	0.0224859	0.0649033	0.1162692	0.1696758
0.4346588	-0.0120681	0.3546498	0.2085517	0.2881891
0.1431172	0.3963426	0.0551770	0.0900927	-0.2332127

Each of the five marginal distributions are univariate normal distributions and this can be seen by overlaying a theoretical normal distribution onto a histogram of the simulated returns for each asset. This is done for Asset 1, AAPL.



Risk Measures

Efficient portfolios can be compared using various risk measures. Volatility, Value at Risk (95%), and Expected Shortfall will be reported on the simulated returns and the efficient portfolios that outperform will be identified.

Volatility

Volatility is the standard deviation of the simulated returns. As a risk measure it is easy to compute, however it captures fluctuations in both directions and does not capture extreme tail risk. Volatility should be prioritised when overall uncertainty is the biggest concern of the investor, in which case, the optimal portfolio would be the GMV portfolio.

Optimal Portfolio using Volatility

```
## Portfolio: GMV Portfolio
```

```
## Weights: 0.03958639 0.1141041 0.255066 0.4184382 0.1728053
```

```
## Theoretical Mean: 0.113884
```

```
## Theoretical Variance: 0.01178675
```

Value at Risk (95%)

Value at Risk (VaR) with a 95% confidence-level is the maximum potential loss over the time horizon with 95% confidence. VaR should be prioritised when the investor faces regulatory or short-term risk limits and the worst-case loss with a given level of confidence must be known. The portfolio that minimises VaR would be optimal.

Optimal Portfolio using VaR

```
## Portfolio: 49
```

```
## Weights: 0.1356872 0.08577409 0.6556939 -0.1873376 0.3101825
```

```
## Theoretical Mean: 0.2640761
```

```
## Theoretical Variance: 0.02768796
```

Expected Shortfall

Expected Shortfall (ES) is the average loss beyond the VaR threshold. ES is the standard risk measure for stress testing and tail risk management, and should be prioritised when the investor is concerned with extreme events beyond VaR. The portfolio that minimises ES would be optimal.

Optimal Portfolio using ES

```
## Portfolio: 40
```

```
## Weights: 0.1003367 0.09619523 0.5083239 0.03549557 0.2596486
```

```
## Theoretical Mean: 0.2088283
```

```
## Theoretical Variance: 0.01814114
```

	Volatility	VaR_95	ES
25	0.1084955	0.0622072	0.1067389
26	0.1087617	0.0564652	0.1008417
27	0.1092702	0.0508717	0.0954918
28	0.1100177	0.0445820	0.0906942
29	0.1109991	0.0394688	0.0863661
30	0.1122085	0.0356078	0.0826036
31	0.1136386	0.0320098	0.0792906
32	0.1152811	0.0293656	0.0764717
33	0.1171270	0.0256883	0.0740692
34	0.1191671	0.0223976	0.0720742
35	0.1213914	0.0191921	0.0704008
36	0.1237900	0.0165431	0.0690077
37	0.1263530	0.0148938	0.0679849
38	0.1290706	0.0128704	0.0672402
39	0.1319333	0.0112378	0.0667768
40	0.1349318	0.0098707	0.0665545
41	0.1380573	0.0080983	0.0666387
42	0.1413013	0.0065576	0.0669771
43	0.1446559	0.0061931	0.0674966
44	0.1481136	0.0058534	0.0682579
45	0.1516673	0.0064160	0.0692787
46	0.1553104	0.0058544	0.0704797
47	0.1590367	0.0055554	0.0718252
48	0.1628407	0.0054021	0.0733414
49	0.1667168	0.0052428	0.0749841
50	0.1706604	0.0059785	0.0767554

Discussion

The analysis shows how portfolio optimisation and risk evaluation can be integrated to inform investment decisions. Through computing historical returns and covariances, an efficient frontier was produced illustrating the tradeoff between expected return and volatility. Possible asset return scenarios were simulated using a Monte Carlo approach under the assumption of multivariate normality, on which the risk measures of volatility, value at risk, and expected shortfall were computed.

Considering the strengths and weakness of these risk measures and comparing them across efficient portfolios yielded important insights, such as that minimising volatility did not minimise tail risk. Portfolios selected on the basis of VaR or ES gave a clear perspective on potential extreme losses, which is important to risk-averse investors.

Conclusion

This project illustrates a practical application of mean-variance portfolio theory and simulation-based market risk analysis. Efficient frontier optimisation selects portfolios with optimal risk-return profiles. VaR and ES give valuable insights into tail risk. Overall, it was demonstrated that this integrated approach to market risk analysis can improve portfolio selection and promote a clearer understanding of practical applications in financial risk management.